Examination Scheme for Semester-VIII

CA – Continuous Assessment EoSE – End of Semester Examination

Regular Students –

Type of Examination	Course Code and Nomenclature	Scheme of Examination	Duration of Examination	Maximum Marks	Minimum Marks
Theory	BTH-88T-851 ENVIRONMENTAL BIOTECHNOLOGY	**CA**	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	BTH-88P-852 ENVIRONMENTAL BIOTECHNOLOGY - PRACTICAL	**CA**	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Theory	BTH-88T-853 GENOMICS AND PROTEOMICS	**CA**	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	BTH-88P-854 GENOMICS AND PROTEOMICS - PRACTICAL	**CA**	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Practical	BTH-88P-855 INDUSTRIAL TRAINING AND PROJECT REPORT -PRACTICAL	**CA**	1 Hrs	30 Marks	12 Marks
		EoSE	03 Hrs	120 Marks	48 Marks

The theory question paper will consist of **two** parts, **A** & **B**.

PART-A: 20 Marks

Part A will be compulsory, having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

PART-B: 60 Marks

Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with an internal choice. Each question will carry 15

Signature of Dean	**Signature of CoC Convenor**	**Signature Of DR (Academic-II)**
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

marks.

*****Evaluation of Industrial Training:** The experimental evaluation based on laboratory work will be of a total of 120 marks, distributed as follows:

(A) 100 Marks — Viva Presentation (by External Examiner)

(B) 20 Marks — Evaluation of Dissertation Report (by External Examiner)

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Raj Jain Dy. Registrar (Academic) University of Rajasthan JAIPUR

UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)								
				Bio-Technology 4th YEAR SEM-VIII	**Credits**			
#	**Level**	**Semester**	**Type**	**Title**	**L**	**T**	**P**	**Total**
7.	8	VIII	MJR	UG0804 -BTH-88T-851 ENVIRONMENTAL BIOTECHNOLOGY	4	0	0	4
8.	8	VIII	MJR	UG0804 -BTH-88P-852 ENVIRONMENTAL BIOTECHNOLOGY - PRACTICAL	0	0	2	2
9.	8	VIII	MJR	UG0804 -BTH-88T-853 GENOMICS AND PROTEOMICS	4	0	0	4
10.	8	VIII	MJR	UG0804 -BTH-88P- 854 GENOMICS AND PROTEOMICS - PRACTICAL	0	0	2	2
11.	8	VIII	MJR	UG0804 -BTH-88P-855 INDUSTRIAL TRAINING AND PROJECT REPORT -PRACTICAL	0	0	6	6

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

BTH-88T-851 ENVIRONMENTAL BIOTECHNOLOGY

Semester	**Code of the Course**	**Title of the Course/Paper**	**NHEQ F Level**	**Credits**
VIII	BTH-88T-851	ENVIRONMENTAL BIOTECHNOLOGY	8	4

Level of Course	**Type of the Course**	**Credit Distribution**			**Offered to NC Student**	**Course Delivery Method**
		Theory	**Practical**	**Total**		
Advanced	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours

List of Programme Codes in which Offered as Minor Discipline	**----------------**
Prerequisites	Intermediate
Course Objectives	➢ To understand the components of the environment and major ecological processes. ➢ To introduce the scope and applications of environmental biotechnology in pollution control, waste management, and environmental sustainability. ➢ To study wastewater and industrial effluent treatment techniques ➢ To explore modern biotechnological tools for pollution detection, environmental monitoring, and to understand related laws, policies.

Detailed Syllabus
BTH-88T-851 ENVIRONMENTAL BIOTECHNOLOGY

Unit-I

Components of the Environment: Hydrosphere, Lithosphere, Atmosphere, Energy transfer in an Ecosystem. **Environmental Biotechnology**: An overview, concept, scope and market, Biological control of air pollution, Bacterial examination of water for potability, Testing of water for physiochemical parameters including BOD & COD, Solid waste: Sources and management (composting, vermicomposting and methane production).

15 Lectures

Unit-II

Waste water: origin, composition and treatment–Physical, chemical and biological treatment of

Signature of Dean	**Signature of CoC Convenor**	**Signature Of DR (Academic-II)**
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

waste water. **Aerobic processes**: activated sludge, oxidation ponds, trickling filter towers, and rotating discs. **Anaerobic processes**: anaerobic digesters, anaerobic filters and up-flow sludge blanket reactors. Microbiology and biochemistry of aerobic and anaerobic waste water treatment processes, Treatment of industrial effluents, and removal of heavy metals from waste waters.

15 Lectures

Unit-III

Kinetic models for biological waste treatment, Bioconversion of agricultural and highly organic waste material into utilizable product, Biogas. **Bioremediation**: Introduction of Bioremediation, Types of bioremediation: Natural (attenuation), Ex-situ and In-situ, Bioaugmentation and bio stimulation. **Biodegradation**: Introduction, Microbial basis of Biodegradation, Biodegradation of Xenobiotics, Microbial degradation of pesticides. **15 Lectures**

Unit-IV

Biotechnological methods of pollution detection: General bioassays in pollution monitoring, cell biology in environmental monitoring, molecular biology in environmental monitoring and biosensors in environmental analysis. Environmental Monitoring (Bioindications, Biomarkers, Biosensors). Environment protection act, environmental Laws and Policies. Waste disposal & Management: Microbiological & biochemical aspects of wastewater treatment process, Microbial strain improvement. **15 Lectures**

BTH-88P-852 ENVIRONMENTAL BIOTECHNOLOGY-PRACTICAL

1. Analyze physico-chemical parameters of water and soil samples.
2. Determine BOD, COD, and DO levels in water samples.
3. Measure TDS, suspended solids, pH, and temperature of water.
4. Estimate acidity, alkalinity, and salinity in water samples.
5. Analyze the hardness and free CO_2 content in water.
6. Determine moisture content, organic matter, and biological parameters in soil samples.
7. Perform MPN and SPC methods for microbiological analysis of water.
8. Study microbial mechanisms involved in the biodegradation of pollutants.
9. Visit and report on processes at an industrial wastewater treatment plant.
10. Any other exercise based on the theory syllabus.

Suggested Books & Text

1. Environmental Biotechnology – Indu Shekhar Thakur
2. Environmental Biotechnology – Bimal C. Bhattacharya & Rintu Banerjee
3. Environmental Biotechnology – R. C. Dubey
4. Biological Process Design for Wastewater Treatment – Larry D. Benefield & Clifford W. Randall
5. Wastewater Engineering: Treatment and Reuse – Metcalf & Eddy (Note: Often published under Tchobanoglous, Burton, and Stensel)

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Raj Jain Dy. Registrar (Academic) University of Rajasthan JAIPUR

6. Environmental Biotechnology – Allan Scragg

Course Learning Outcomes:

By the end of the course, students will be able to:

1. Understand and explain the basic components of the environment and the principles of energy flow.
2. Apply biotechnological approaches to monitor, manage, and treat environmental pollution, including the biological treatment of air, water, and solid waste.
3. Analyse wastewater composition and treatment processes, including physical, chemical, aerobic, and anaerobic methods, with an understanding of their microbial and biochemical foundations.
4. Evaluate and implement bioremediation and biodegradation techniques for the treatment of pollutants and hazardous compounds using microbial systems.
5. Utilize modern biotechnology tools and legal frameworks for environmental monitoring, biosensor application, and the development of eco-friendly waste management strategies.

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

BTH-88T-853 Genomics and Proteomics

Semester	Code of the Course	Title of the Course/Paper	NHEQF Level	Credits
VIII	BTH-88T-853	GENOMICS AND PROTEOMICS	8	4

Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Advanced	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours

List of Programme Codes in which Offered as Minor Discipline	----------------
Prerequisites	Intermediate
Course Objectives	➢ To understand the organization, structure, and sequencing of genomes in prokaryotes, eukaryotes, and viruses. ➢ To learn various gene identification techniques, genetic mapping tools, and expression analysis methods. ➢ To explore applications of functional genomics and gene editing technologies like CRISPR-Cas9. ➢ To gain knowledge of protein structure, synthesis, and analytical techniques in proteomics.

Detailed Syllabus

BTH-88T-853 Genomics and Proteomics

Unit-I

Introduction - Organization and structure of genomes (prokaryotic, eukaryotic, viral), concept of gene, genome size, introns & exons, sequence complexity. Genome structure in viruses and prokaryotes. Chromosome isolation, microdissection, and retrofitting. Genes and proteins, types of polymorphisms, genome databases, and gene discovery. DNA extraction and genomic DNA preparation. DNA sequencing methods: Maxam-Gilbert, Sanger, Fluorescence, Shotgun, NGS techniques (overview and principles). **15 Lectures**

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Rj Jain Dy. Registrar (Academic) University of Rajasthan JAIPUR

Unit-II

Gene Identification and Expression – Inheritance patterns, mutations, Single Nucleotide Polymorphisms (SNPs), Expressed Sequenced Tags (ESTs), gene-disease links, diagnostic genes, drug targets. Genome projects: *E. coli, Arabidopsis*, rice, Human Genome Project. Genome annotation, gene prediction software, gene function identification, and gene ontology. Comparative & structural genomics. Model systems: *Drosophila*, Yeast, *C. elegans*. Gene expression profiling (traditional and RNA-based), applications in medicine, and gene knockdown.

15 Lectures

Unit-III

Introduction to Proteins – Protein and peptide preparation, Merrifield synthesis, peptides as probes, proteins as drugs. Two-hybrid interaction screens. Mass spectrometry in protein expression analysis. SDS-PAGE, 2D-PAGE, protein cleavage, Edman microsequencing. Automation in proteomics. Proteome mining, linking genomics and proteomics. Applications in drug discovery, toxicology, and the use of phage antibodies. **15 Lectures**

Unit-IV

Analysis of Proteomes – 2D-PAGE: sample preparation, solubilization, reproducibility. Protein detection in gels, image analysis. Mass spectrometry for protein identification: de novo sequencing, correlation-based methods. 2D-electrophoresis with mass spectrometry. Microarray techniques – types, experiment design, disease treatment applications. Genotyping tools – DNA chips, diagnostic assays, and services. **15 Lectures**

BTH-88P-854 GENOMICS AND PROTEOMICS-PRACTICAL

1. Introduction to genome analysis techniques.
2. Linkage analysis and gene mapping problems in *Drosophila*.
3. Linkage and chromosomal mapping problems in *Neurospora*.
4. Exercise based on Pedigree analysis problems in humans.
5. Protein separation using SDS-PAGE.
6. Study of protein sequence databases.
7. Sequence retrieval and alignment methods (demonstration & exercises).
8. Practicals based on theoretical concepts.

Suggested Books & Text

1. S. B. Primrose & R. M. Twyman – *Principles of Genome Analysis and Genomics*, 7th Edition, Blackwell Publishing, 2006.
2. S. Sahai – *Genomics and Proteomics: Functional and Computational Aspects*, Plenum Publishing, 1999.
3. Andrezej K. Konopka & James C. Crabbe – *Compact Handbook of Computational Biology*, Marcel Dekker, USA, 2004.
4. Pennington & Dunn – *Proteomics: From Protein Sequence to Function*, 1st Edition, Academic Press, San Diego, 1996.

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

5. Arthur M. Lesk – *Introduction to Genomics*, Oxford University Press, 2007.
6. A.M. Campbell & L. J. Heyer – *Discovering Genomics, Proteomics and Bioinformatics*, Pearson Education, 2007.

Course Learning Outcomes:
By the end of the course, students will be able to:
1. Students will be able to explain the structure, organization, and sequencing methods of genomes in prokaryotes, eukaryotes, and viruses.
2. They will develop the ability to identify genes, analyse genetic variations, and apply tools for genetic mapping and genome annotation.
3. Learners will gain insights into functional genomics, and gene expression profiling
4. They will understand protein structure, synthesis, and various techniques used in proteomic analysis, including SDS-PAGE and mass spectrometry.
5. Students will be able to integrate genomic and proteomic knowledge for applications in biomedical research, diagnostics, and drug development.

BTH-88P-855 INDUSTRIAL TRAINING AND PROJECT REPORT

<table>
<tr><th>Semester</th><th>Code of the Course</th><th colspan="4">Title of the Course/Paper</th><th>NHEQF Level</th><th>Credits</th></tr>
<tr><td>VIII</td><td>BTH-88P-855</td><td colspan="4">INDUSTRIAL TRAINING AND PROJECT REPORT- PRACTICAL</td><td>8</td><td>6</td></tr>
<tr><th rowspan="2">Level of Course</th><th rowspan="2">Type of the Course</th><th colspan="3">Credit Distribution</th><th rowspan="2">Offered to NC Student</th><th colspan="2" rowspan="2">Course Delivery Method</th></tr>
<tr><th>Industrial training</th><th>Project report</th><th>Total</th></tr>
<tr><td>Adavnced</td><td>Major</td><td>4</td><td>2</td><td>6</td><td>NO</td><td colspan="2">Industrial Training on a relevant syllabus topic</td></tr>
<tr><th colspan="2">List of Programme Codes in which Offered as Minor Discipline</th><td colspan="6">----------------</td></tr>
<tr><th colspan="2">Prerequisites</th><td colspan="6">Intermediate</td></tr>
<tr><th colspan="2">Course Objectives</th><td colspan="6">➢ To provide students with comprehensive exposure to industrial operations and professional environments.
➢ To facilitate the integration of theoretical knowledge with practical industrial applications.
➢ To enhance students' technical proficiency and capacity for analytical problem-solving.</td></tr>
</table>

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

Rekha

	➢ To foster the development of professional competencies, including communication, collaboration, and ethical conduct.

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

Report of Industrial Training of 60 days:

As part of the **Industrial Training program**, students are expected to gain practical experience in **industry-oriented, clinical, or production environments**, at recognized industrial/training organizations/company/universities equipped with advanced facilities. The main objective of this training is to enhance students' **employability and technical competence** by exposing them to **real-time industrial processes, workflows, or services**. Students must choose a topic or domain relevant to their field that adds **practical knowledge and skill-based learning**, under the supervision of an assigned **faculty guide or industrial trainer**. They are required to understand operational systems, observe industrial standards, participate in hands-on technical tasks (such as production, quality control, diagnostics, or instrumentation), and document the practical exposure. At the conclusion of training, each student must prepare a well-structured **Industrial Training Report** in **Times New Roman, font size 12**, which should include:

(i) **Title page-** Report title, name of student, name of organisation, duration (start and end dates)
(ii) **Certificate/Declaration-** from training organisation stating that you completed the training.
(iii) Acknowledgement
(iv) Abstract/summary
(v) Table of contents
(vi) **Introduction** – describing the industrial organization, nature of work, and background of the training;
(vii) Organisation Profile: name, background, products/services offered, organisation structure
(viii) Training Details

a) Work description
b) Techniques and procedures learned
c) Observation and Results- with Tables, graphs and photographs
d) Discussion
e) Skill gained and Learning Outcomes
f) Challenges faced and solutions
g) Conclusion
h) References
i) Annexures- if any (Raw data, SOPs etc.)

- The final report should showcase the student's **professional understanding, industrial exposure, and application of academic knowledge to real-world scenarios**, and will serve as an essential academic and career document.

Evaluation of Industrial Training
The experimental evaluation based on laboratory work will be of **120 marks (EOSE)**, distributed as follows:
(A) 100 Marks — Viva Presentation (by External Examiner)
(B) 20 Marks — Evaluation of Training Report (by External Examiner)

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR

Evaluation Component	INDUSTRIAL TRAINING AND PROJECT REPORT (with proper certification)
Evaluation Method	Appropriate to the material and specific content
Learning Outcome	Students will demonstrate the ability to work at a company or governmental organization

Course Learning Outcomes:
By the end of the course, students will be able to:

1. Students will demonstrate comprehensive proficiency in fundamental and advanced methodologies utilized in biotechnology.
2. Students will exhibit the capability to operate standard and sophisticated laboratory instrumentation, analytical devices, and computational tools in a safe and effective manner.
3. Students will apply independent, analytical, and critical thinking skills in conjunction with the effective utilization of diverse informational and research resources.
4. Students will articulate a thorough understanding of the principles, mechanisms, and interrelationships governing *in vivo* and *in vitro* biochemical, molecular, and genetic processes.

Signature of Dean	Signature of CoC Convenor	Signature Of DR (Academic-II)
	Rekha	Dy. Registrar (Academic) University of Rajasthan JAIPUR